

FULTON COUNTY AP, IN, USA

WMO: 720736

Lat: 41.066N

Lon: 86.182W

Elev: 241

StdP: 98.46

Time Zone: -5.00 (NAE)

Period: 10-19

WBAN: 00265

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.9	-13.0	-21.4	0.6	-15.4	-18.0	0.8	-12.7	10.4	0.9	9.3	-2.8	2.6	250	0.472

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	33.0	23.4	32.3	23.2	31.1	22.6	25.7	30.7	24.9	29.9	24.0	28.5	4.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.6	27.5	23.0	18.2	26.9	22.6	17.8	26.5	80.2	31.0	76.5	30.2	73.6	28.8	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.1	8.1	7.3	-22.0	36.1	3.1	1.9	-24.2	37.5	-26.1	38.6	-27.8	39.6	-30.1	41.0
			-22.3	26.7	3.1	0.9	-24.6	27.3	-26.4	27.8	-28.1	28.3	-30.4	28.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.8	-2.3	-0.7	4.7	10.8	18.1	22.5	24.4	23.1	19.8	13.0	5.8	1.3		
	DBStd	10.64	6.34	6.61	6.05	4.70	5.21	3.36	3.30	2.74	4.44	4.93	5.18	5.54		
	HDD10.0	1368	383	303	188	44	3	0	0	0	0	23	147	275		
	HDD18.3	3027	641	532	425	228	71	7	1	2	34	179	378	529		
	CDD10.0	2015	1	4	24	69	255	376	447	406	293	116	19	4		
	CDD18.3	632	0	0	2	3	64	132	189	149	78	13	1	0		
	CDH23.3	5832	0	0	14	40	608	1136	1921	1300	694	118	2	0		
	CDH26.7	2041	0	0	1	3	196	394	771	414	237	25	0	0		
(14) Wind	WSAvg	3.1	3.8	3.7	3.7	3.9	3.2	2.7	2.1	2.0	2.4	3.1	3.5	3.5		
(15) Precipitation	PrecAvg	974	52	44	65	96	104	102	107	92	89	72	85	68		
	PrecMax	1578	155	153	139	209	186	245	244	315	252	228	230	202		
	PrecMin	687	0	0	11	13	21	34	19	16	1	7	25	8		
	PrecStd	172	35	31	29	43	42	45	62	59	55	45	43	38		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	17.9	26.3	27.2	33.7	34.9	37.1	33.0	35.0	29.9	22.5	17.4		
		MCWB	12.9	12.9	17.8	16.4	20.3	23.2	24.7	23.7	23.1	20.8	15.5	14.5		
	2%	DB	12.1	15.2	21.1	24.7	30.9	32.5	33.7	32.2	32.2	27.0	18.8	14.8		
		MCWB	10.3	12.0	14.6	15.9	20.3	22.5	24.3	23.6	23.0	18.7	14.6	12.4		
	5%	DB	8.8	12.0	17.4	22.2	28.8	31.1	32.4	30.9	29.8	23.9	16.3	11.1		
		MCWB	7.2	9.4	13.0	14.9	19.3	21.6	23.9	23.1	21.7	17.3	12.2	9.4		
	10%	DB	6.1	7.6	14.6	19.2	27.1	28.9	31.1	28.9	27.3	21.4	13.0	7.9		
		MCWB	4.2	5.1	10.3	13.2	18.7	21.1	23.3	22.3	20.0	16.5	9.8	6.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.1	15.3	17.9	19.1	23.0	25.5	27.0	26.9	25.1	22.1	17.8	15.9		
		MCDB	14.4	17.1	25.3	24.1	28.7	31.3	32.1	28.6	31.3	27.2	20.5	16.8		
	2%	WB	10.3	11.9	15.9	17.4	21.9	24.5	26.0	25.5	23.8	20.6	15.7	12.4		
		MCDB	11.3	14.4	20.3	21.1	27.1	30.0	31.5	29.3	28.9	24.3	17.8	13.9		
	5%	WB	7.1	8.4	13.6	15.9	20.8	23.4	25.1	24.5	23.0	19.0	12.8	9.1		
		MCDB	8.0	11.5	16.5	20.1	25.9	28.1	30.4	28.5	27.5	22.2	15.9	10.7		
	10%	WB	4.3	5.3	10.9	14.5	20.0	22.5	24.3	23.6	22.1	17.2	10.4	7.1		
		MCDB	6.2	7.5	14.1	19.3	25.2	26.9	29.0	27.3	26.2	20.5	13.6	8.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.5	8.0	9.1	11.2	11.1	10.5	10.8	11.0	11.3	10.7	8.4	6.5		
		MCDBR	10.1	11.7	12.6	14.5	13.2	12.5	11.6	12.2	12.4	13.3	12.1	9.6		
	5% WB	MCWBR	8.6	9.4	8.3	8.5	5.8	5.0	4.7	5.2	4.8	6.7	8.1	8.2		
		MCWBR	9.3	11.3	11.5	12.5	10.7	9.9	10.2	9.0	10.2	10.7	10.9	8.9		
(40) Clear-Sky Solar Irradiance	taub	taub	0.295	0.324	0.346	0.407	0.444	0.436	0.449	0.450	0.417	0.355	0.318	0.299		
		taud	2.417	2.327	2.362	2.202	2.148	2.216	2.181	2.167	2.239	2.416	2.464	2.441		
	Ebn at Noon	Ebn at Noon	863	882	902	864	838	843	827	813	816	838	828	826		
		Edh at Noon	80	101	110	139	150	141	144	142	124	93	76	72		
(44) All-Sky Solar Radiation	RadAvg		1.59	2.44	3.58	4.66	5.33	6.07	6.09	5.37	4.42	2.94	1.88	1.33		
	RadStd		0.15	0.29	0.29	0.42	0.49	0.39	0.34	0.31	0.36	0.29	0.24	0.14		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (48 neighbors)		(m)								
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page